

AN4078_C

Author: Anand Srinivasan

Associated Project: No

Associated Part Family: CY7C68013A/14A/15A/16A

Software Version: None

Associated Application Notes: None

Application Note Abstract

The EZ-USB FX2LP™ is a next-generation USB high-speed controller. EZ-USB FX2LP enhances the functionality of the EZ-USB FX2™ while minimally effecting existing designs. This application note serves two purposes:

1. Assist in migrating existing EZ-USB FX2 (also referred to as FX2 in this application note) applications to EZ-USB FX2LP (also referred to as FX2LP in this application note).
2. Assist the designers familiar with FX2 in creating FX2LP based applications.

This application note highlights the differences between the two products. It also provides a brief description of the whole product support collateral available for development work with FX2LP.

Introduction

This application note provides a summary of the main items to consider while replacing the FX2 in an existing application with an FX2LP part. The changes are categorized into required essential changes for all FX2LP applications migrating from FX2 and changes and other applications based changes that “may” be required in your application due to additional enhanced feature in FX2LP.

Essential Changes

Following are a list of required changes when replacing the FX2 in a existing design with FX2LP:

- Make sure to replace the crystal with the appropriate load capacitors. The FX2LP requires a crystal with load capacitors of 12 pF.
- Make sure that the reset is held for at least 5 ms.
- Make sure that the V_{CC} ramp-up time is at least 200 μ s with a ramp rate of 18V per millisecond.
- Changes on AVCC/AGND pin functionality (see [Changes on AVCC/AGND Pin Functionality on page 3](#)).

Note After installing [CY3684 EZ-USB FX2LP Development Kit](#), the hardware collateral of the kit available at location C:\Cypress\USB\Hardware (Location varies based on installation) can be used as a reference to develop the board.

Application Based Changes

As next generation of the FX2, the FX2LP has some enhanced features that the user may want to take advantage of or may decide to simply disregard. Based on the application's needs these enhancements in the FX2LP may require some changes in the designer's FX2-based application. Following is a list of things that the designer needs to be informed of when replacing the FX2 in a existing design with FX2LP.

- Is the existing FX2 application bus-powered and has the double enumeration workaround for meeting the unconfigured current limit of 100 mA? If so, set the control bit of the EEPROM to 0 (chirp enabled) as the workaround is not required anymore. The FX2LP meets the 100-mA current requirement for unconfigured current. You may opt to either leave in or remove the double enumeration workaround in the firmware.
- The FX2LP automatically does a disconnect and a subsequent reconnect on a hard reset (asserting RESET# pin). If the FX2 firmware has the 8051 do this disconnect and reconnect manually by setting and clearing DISCON bit of the USBCS register on a hard reset, it is not required anymore. Based on your application's needs, when porting the same firmware to FX2LP, you may opt to either leave in or remove the code that does this disconnect and reconnect manually. This needs to be evaluated on an application basis and the effects (if any) on the system as a whole on disconnecting and reconnecting twice (if the 8051 code is left in the firmware ported to FX2LP).
- FX2LP has internal memory from 0x0000 to 0x3FFFF as opposed to FX2 which has internal memory from 0x0000

to 0x1FFFF. Check the code location to make sure whether the existing FX2 application uses memory mapped ROM (any nonvolatile memory) or memory mapped I/O within the address space of 0x2000 to 0x3FFF. If so, they must be mapped to new locations and the firmware should be relinked as the FX2LP accesses the internal memory which is not the intended access method of the end application. In the limited number of designs that require this change, just a logic change to an FPGA or programmable logic array for hardware memory decoding and changing the target location of the external code within the compiler/linker is needed. If the existing FX2 design has RAM at this location (0x2000 to 0x3FFF), no changes are required while replacing the part with FX2LP.

- Does the FX2 application use high-bandwidth Isochronous IN transfer? If so, does it have a workaround (on the host end or the device end) for handling the possible occurrence of a scenario of data PID mismatch: additional hardware, external logic, or software on the host end to handle the scenario? If so, you may consider removing the additional logic/code as the FX2LP has resolved this data PID mismatch issue. You may also choose to not remove the designed workaround for ease of drop-in replacement with FX2LP

Note While changing components such as the USB controller (FX2 to FX2LP), the USB-IF requires a rerun of certification test. The following section provides details on the specifications on every change that is mentioned earlier.

Details of the Essential Changes

An overview of the essential hardware changes/additions in the FX2LP are listed below:

- Changes to Crystal Specification
- Minimum Reset Time
- V_{CC} Ramp Up Timer
- Changes to AVCC/AGND Pin Functionality

Changes to Crystal Specification

With the use of FX2LP one change is required. When a crystal is used in the design, for proper operation the load capacitance of the crystal must change. This affects both the load capacitors and the crystal being used. The CY7C68013 requires a crystal with load capacitors that are between 20 pF and 33 pF. The CY7C68013A requires a crystal with load capacitors of 12 pF. If redesigning an FX2 application with FX2LP, the crystal must change to a load capacitance of 12 pF, and the matching load capacitors must change to 12 pF. This is a requirement of all designs using FX2LP. Following is the spec on the crystal:

- 24 MHz ± 100 ppm
- Parallel resonant
- Fundamental mode

- 500 μ W drive level
- 12-pF (5% tolerance) load cap

One of the parts we recommend is one from ECS Inc. International: ECS-240-20-4.

The above specification must be taken into consideration while selecting both the load capacitors and the crystal. Using a different crystal load capacitance with a crystal specified for 12 pF is expected to have some effect on the frequency shift. It is recommended that the designer always make sure that the power dissipated by the crystal is within the crystal manufacturer's specifications. Over-driving the crystal may damage the crystal. Refer to the specific crystal manufacturer's recommendations.

An article on crystal design and selection from [Ecliptek's](#) sitestates the following:

"The rate of aging is typically greatest during the first 30 to 60 days after which time the aging rate decreases. The following factors effect crystal aging: adsorption and desorption of contamination on the surfaces of the quartz, stress relief of the mounting and bonding structures, material outgassing, and seal integrity."

One of the effects of over-driving the crystal out of its specified conditions may be aging of the crystal, which is expected to result in frequency shift. From speaking to various crystal manufacturers, we have been informed that the majority of the shift due to aging usually occurs in the first 45–60 days or so.

In general, it is recommended that a designer always make sure that the power dissipated by the crystal is within the crystal manufacturer's specifications. Over-driving the crystal 'may' have adverse effects on the crystal performance and accuracy. It is also recommended that the user refer to the crystal manufacturer's recommendations on the various effects of driving a crystal beyond its specified values.

There have been reports from few FX2LP customers that the use of the same crystal with 16-pF or 20-pF load capacitance does not seem to cause any problems. Even though the crystal with load capacitance of 16 pF or 20 pF appears to work (and probably works for a long time) we are unable to provide test reports or any test data on this, as the part was designed and tested to use the crystal as specified earlier.

Using a 16-pF or 20-pF crystal or any other standard load capacitance, may work as indicated by the discussions we have had with various crystal manufacturers. But using this is at the customer's discretion as the FX2LP part was only tested and designed for a 12-pF, 24-MHz crystal. Please consult with the specific crystal manufacturer for information on the various effects of driving a crystal beyond its specified values. When a crystal is used in the design using FX2LP, Cypress recommends that for proper operation the load capacitance of the crystal should be 12 pF.

Minimum Reset Time

The minimum reset time spec'ed for the FX2LP part is 5 ms. The FX2 was designed with a minimum reset time of 1.91 ms

V_{CC} Ramp Up Time

To obtain the power savings, the FX2LP parts use a core which is powered from 1.8 Volts. To maintain common power sources with the FX2LP silicon, this voltage is derived from the 3.3V input. To properly power the internal regulator the 3.3V input must have a limit on the power ramp-up rate. This means the input must turn on to a valid voltage (3.0V to 3.6V) over a 200-μs period. This equates to an 18V-per-millisecond ramp rate. Powering up the input too quickly latches the internal regulator and cause problems in powering up the core. Many 3.3V regulators have turn-on times that are long enough to properly power-up the internal regulator. One such regulator is the LT1763CS8-3.3.

Changes on AVCC/AGND Pin Functionality

As far as PCB changes are concerned, there are no major ones. Both FX2 and FX2LP run on a single 3.3V supply. There are two additional AVCC/AGND pins on the FX2LP that can be rerouted on the PCB to provide additional filtering. These two additional pins exist in the 56-, 100-, and 128-pin packages.

For the 128-pin TQFP package, AVCC is pin 17 (was VCC on the FX2), and AGND is pin 20 (was GND on the FX2). Pin 17 of the FX2LP128 pin package does not need a duplicate of the pin 10 AVCC filter network. It can be wired in parallel to pin 10.

The FX2LP can use the same filter network as the FX2 for AVCC. The FX2LP analog ground (pin 20 in 128 pin package) does not need to be isolated. It can be connected to the digital ground. Other changes made to the chip do not affect all designs.

Details of Application-based Changes

This section highlights major improvements and changes in FX2LP that may affect the older FX2 design while migrating to FX2LP. Most of these do not affect your design. The information given here is the evaluation on how it may effect your application design. An overview of these features are listed as follows:

- Disabling high-speed chirp via EEPROM config byte
- Automatic Disconnect and reconnect on a hard reset
- Expanded internal code/data RAM
- Zero-length in-packets with no firmware intervention and Data PID sequencing in isochronous IN transfers

Disabling High-speed Chirp via EEPROM Config Byte

During power-on sequence, the operating speed of the FX2 device defaults to high-speed. The FX2 device has the capability of disabling the high-speed chirp state machine using the Cypress internal register CT1. The 8051 can set or clear bit 1 of the CT1 register anytime. On re-enumeration (disconnecting and then reconnecting) the FX2 device enumerates with the chirp state machine disabled if this bit of the CT1 register is set to 1. Therefore, on a disconnect event followed by a re-connect, the device re-enumerates as a full-speed only device.

The chirp state machine can also be disabled on power-up, by setting bit 7 of the configuration byte of the EEPROM. During power-up sequence, the core copies this bit setting to bit 1 of the CT1 register and the device enumerates as a full-speed device. After enumerating as a full-speed device, the 8051 may enable the chirp state machine anytime by clearing bit 1 of the CT1 register. On re-enumeration the same device enumerates as a high-speed device.

In the case of FX2LP, the device also defaults to a high-speed mode of operation: bit 1 of the CT1 register defaults to 0. After the device has powered-up with the chirp state machine enabled (default), it can be switched to a different mode of operation by having the 8051 change bit 1 of the CT1 register.

If no EEPROM is used, the device operates in its default high-speed mode. If using an EEPROM, the chirp state machine status is determined by the setting of bit 7 of the configuration byte of the EEPROM. The behavior of the FX2LP chirp state machine is different than the FX2 when bit 7 of the configuration byte of the EEPROM is set to 1 (chirp state disabled on power-up).

The CT1 register and the EEPROM control bit (bit 7 of the configuration byte) behave differently in FX2LP than in FX2. In FX2, if the EEPROM control bit that forces USB full-speed operation is set, FX2 comes up in full-speed mode, and upon re-enumeration FX2 is capable of re-enumerating on the USB bus in high-speed mode (if the 8051 has cleared bit 1 of the CT1 register to enable high-speed mode). In the case of FX2LP, if the full-speed control bit of the EEPROM is set, the device ONLY comes up in full speed mode, even after USB re-enumeration. Setting or clearing this bit by the 8051 has no effect on enabling or disabling the chirp state machine when bit 7 of the configuration byte of the EEPROM is set to 1. If it is set to 0, FX2LP behaves the same as the FX2. In other words when bit 7 of configuration byte is set to 0 (chirp enabled) the FX2LP comes up in high-speed mode and the 8051 may switch the mode by changing bit 1 of CT1 register anytime followed by a disconnect and reconnect event.

If the existing FX2 application is bus powered and uses the double enumeration sequence (which is a recommended workaround) to meet the unconfigured current requirements of the USB-IF compliance test, then the application does not

behave the same when replaced with the FX2LP. If replacing the part with FX2LP, when the device enumerates with the high-speed chirp disabled (control bit of the configuration byte of the EEPROM set), on reenumeration, the device fails to enumerate as a high-speed device even though the 8051 had cleared bit 1 of the CT1 register. The full-speed mode trick upon initial plugin FX2 was done to get the unconfigured power-down to the 100-mA range to meet the USB specification. As FX2LP power is 50–60 mA maximum, not meeting the unconfigured current limit when using the device as bus powered is no longer an issue—there is no need to set this control bit in the EEPROM anymore. The workaround is not required anymore. Just have the FX2LP application enumerate in USB high speed mode from the start.

For an FX2 bus-powered application that uses double enumeration sequence to behave the same when replaced with FX2LP, all you need to do is set the control bit of the EEPROM to 0 (chirp enabled) and can leave the firmware 'as is' with the double enumeration workaround code in it.

Automatic Disconnect and Reconnect on Hard Reset

The FX2LP has an enhanced capability of disconnecting and reconnecting on a hard reset, when the RESET# is asserted by the external peripheral.

The FX2 does not cause a disconnect on a hard reset (RESET# asserted). On releasing the reset the FX2 no longer responds to the host since it does not have a valid address anymore. To resume communication with the host the 8051 must cause a disconnect and a reconnect by setting DISCON bit of the USBCS register and subsequently clearing. On a reconnect event the host issues a SET_ADDRESS request and re-enumerate the device.

In the case of an FX2LP, the device does a disconnect and a reconnect automatically on a hard reset. When the RESET# is asserted, the FX2LP resets the FNADDR (function address) register to 0x00. On releasing the reset# the FX2LP automatically does a disconnect followed by a reconnect. If using FX2LP the 8051 does not need to be programmed to do the disconnect and reconnect by setting and clearing the DISCON bit manually.

Expanded Code/Data RAM

The FX2LP has 16 Kbytes of internal Code/Data RAM, where the FX2 had only 8 Kbytes. The additional RAM is located in the address space of 0x2000 to 0x3FFF. If the firmware of the existing design is completely internal to the FX2, no changes are required to use the FX2LP. If the FX2 design had RAM at this location, no changes are required. The FX2LP accesses the internal RAM instead of the external RAM.

If there was either memory mapped ROM (any nonvolatile memory) or memory mapped I/O within these locations, they must be mapped to new locations and the firmware should be relinked. In the limited number of designs that require this change, just a logic change to a FPGA or programmable logic array for hardware memory decoding and changing the target location of the external code within the compiler/linker is required.

In most designs it is felt that the external memory map would not have used this location in memory. Less logic is required to locate the memory at higher locations and therefore it is believed that most designs would have used a higher address, such as 0x8000, to start the external memory. If this is the case, no modifications, other than the required crystal modifications, are required.

Zero-length INPackets with No Firmware Intervention and Data PID Sequencing for ISO Transfers

The FX2LP has the capability of sending a zero-length isochronous data packet (ZLP) when the host issues an IN token to an isochronous IN endpoint FIFO and the SIE does not have any data packets available.

This feature is very useful when designing high-bandwidth isochronous applications. When an isochronous IN endpoint is configured for greater than one packet per microframe, there is a possibility of the core not having more than one packet available in a microframe. In this case, when the host issues an IN token, the FX2LP core sends a zero-length packet with the appropriate data PID automatically. Hence avoiding the occurrence of a scenario where the host may encounter a turnaround time-out error on not receiving any data when requesting more than one packet per microframe.

In the Technical Reference Manual, register EPxISOINPKTS (x = Endpoint number) defines an additional bit called: ADDJ. This bit defaults to a zero value. In this condition, FX2LP operates the same as the FX2.

The auto adjust (AADJ) feature was useful when designing with the FX2LP engineering samples provided at the early stages of the FX2LP development. These engineering samples did not have the ability to issue a zero-length isochronous packets automatically and hence were prone to running into a DATA PID mismatch scenario when dealing with high-bandwidth ISO IN transfer (please refer to the “*Streaming Data Through Isochronous/Bulk Endpoints on EZ-USB FX2™ EZ-USB FX2LP™ - AN4053*” application note for further information on DATA PID mismatch issue). The auto adjust feature is a workaround to the data PID mismatch issue and to use this workaround, the wMaxPacketSize that could be defined in the isochronous endpoint descriptor is limited to 1024 bytes.

Feature Enhancements

This section is for designers already familiar with the FX2 architecture who are creating a new application using FX2LP. This section highlights additional (along with the essential and application based changes mention in the prior two sections) improvements and changes in FX2LP. An overview of the feature changes/additions in the FX2LP are listed as follows:

- Lower power
- Enabling pull-up on D+
- FIFO Address Lines Hold Time for Asynchronous Write
- ECC generation on GPIF data
- Additional Package with more GPIOs (CY7C68015A/16A).

Lower Power

As far as the power source is concerned, the FX2LP uses a different process than the FX2. The FX2LP uses an internal 1.8 V regulator. The major result of this change is to reduce the power consumption of the chip. The lower current is obtained just by using the FX2LP with the required crystal modifications. This configuration meets all USB bus powered requirements of unconfigured current (100 mA) and suspend current (500 μ A).

As a result of this lower power, to meet the unconfigured current limit, it is no longer necessary to use the double enumeration procedure where the device first enumerates in full-speed with a current draw below 100 mA and then disconnects and reconnects as a high-speed device. If an existing design uses this procedure the designer can remove the procedure in design. For new designs using FX2LP, there is no concern as the part does meet the 100-mA limitation requirement for unconfigured current while operating in either full or high speed.

Enabling Pull-up on D+

There is a minor difference with respect to when the internal logic enables the pull-up on D+ to signal an attach event to the host. In the case of the FX2, the pull-up on D+ is enabled on power-up. In the case of the FX2LP, the pull-up on D+ is enabled when the reset is released (in a deasserted state).

On plug-in, the FX2 enables the pull-up on D+ signaling at attach event to the host. If the reset is held active for longer than 100 ms, an attach signal is sent to the host and the host begins its enumeration sequence (after 100 ms from when the attach is detected), while the FX2 is still held in reset. Section 7.1.7.3 (page 150) of the USB 2.0 specification, provides further information on the debounce interval (Δt_3). While using the FX2 Cypress recommends that a reset time of 10 ms with an RC network of 100K/0.1 μ F be used.

The FX2LP has a more flexible reset timing requirement. As the FX2LP enables the pull-up on the D+ after the RESET is deasserted, there really is no restriction on the reset timing as only the host receives an attach signal when the RESET is released.

FIFO Address Lines Hold Time for Asynchronous Write

In the case of an FX2 device, when operating in slave FIFO mode in an asynchronous interface, the minimum hold time (t_{FAH}) for FIFO address lines (FIFOADR[1:0]) for a slave FIFO write operation is 70 ns, which is measured from the deasserting edge of SLWR.

This has been improved and optimized to 10 ns when using the FX2LP parts. The FIFOADDR[1:0] lines hold time (t_{FAH}) for a slave FIFO asynchronous write is spec'ed for a minimum of 10 ns for the FX2LP.

ECC Generation on GPIF data

This is a new feature and does not affect existing designs. If ECC generation is to be added to an existing design there are additional registers that enable this function.

The FX2LP can be configured to either calculate two separate 256-byte ECCs on two consecutive 256-byte blocks of data, or alternatively one single 512-byte ECC on a 512-byte block of data. After the 8051 resets the ECC calculation by writing any value to ECCRESET, the FX2LP calculates ECC on any data bytes that transfer across the GPIF or Slave FIFO interface. The FX2LP stops ECC calculation after all 512 bytes have been processed and it waits for a new ECC reset from the 8051 before it commences any new calculations.

The additional registers for this function are:

ECCCFG:	Configuration register (256/512)
ECCRESET	Reset ECC-byte registers to zero
ECC1B0,	This is the second eight bits of the line parity on the first 256-byte block or the 512-byte block.
ECC1B1	Lower eight bits of line parity on first 256-byte block or lower eight bits of line parity on 512-byte block.
ECC1B2	This is the 6-bit column parity on first 256-byte block or 6-bit column parity on and upper two bits of line parity for 512-byte block.
ECC2B0	This is the second eight bits of the line parity. on the second 256-byte block
ECC2B1	Lower eight bits of line parity on second 256-byte block
ECC2B2	This is the 6-bit column parity on the second 256-byte block

(For column and line parity see Smartmedia Specification.)
See Technical Reference Manual for check/correct sample code.

Package Changes

Additional options have been made available in the packaging of the FX2LP part. Additional configurations are available in the 56-pin packages. These configurations allow for two additional GPIO pins. These parts are the part number CY7C68015A/CY7C68016A, available in both a QFN and a SSOP package. In CY7C68015A/16A, IFCLK and CLKOUT pins are replaced by PE0 and PE1 respectively. The effect of this change on the design has to be considered while replacing FX2 (56-pin QFN package) with CY7C68015A/16A. As in FX2, all FX2LP parts are available in lead-free packages. The following table lists the FX2 parts and the respective FX2LP part that may be used to replace the FX2. Besides the packages that are specified, FX2LP also has a VFBGA 56-pin package (CY7C68013A-56BAXC and CY7C68014A-56BAXC).

CY7C68014A and CY7C68016A are Ideal for battery-powered applications as they have an ultra-low suspend current of 100 μ A (typical). CY7C68013A and CY7C68015A, which have a low suspend current of 300 μ A (typical), are ideal for non-battery powered (self or bus powered) applications.

EZ-USB FX2 Part Number	EZ-USB FX2LP Part Number	Package Description
CY7C68013-56PVC or CY7C68013- 56PVXC	CY7C68013A-56PVXC or CY7C68013A-56PVXI or CY7C68014A-56PVXC	56-pin SSOP
CY7C68013-56LFC or CY7C68013- 56LFXC	CY7C68013A-56LTXC or CY7C68013A-56LTXCT or CY7C68013A-56LTXI or CY7C68014A-56LTXC	56-pin QFN
CY7C68013-100AC or CY7C68013- 100AXC	CY7C68013A-100AXC or CY7C68013A-100AXI or CY7C68014A-100AXC	100-pin TQFP
CY7C68013-128AC or CY7C68013- 128AXC	CY7C68013A-128AXC or CY7C68013A-128AXI or CY7C68014A-128AXC	128-pin TQFP

EZ-USB FX2LP whole Product Support Collateral

Development Tool

The EZ-USB FX2LP has a development kit similar to that of the EX-USB FX2. The part number for this kit is CY3684. The development kit for the FX2LP contains a development board, cables, documentation and a development CD.

The CD has electronic copies of the Technical Reference Manual, the data sheet, schematics and gerbers for the development board, firmware examples, drivers (CyUSB.sys) and evaluation version of the Keil's μ Vision2 Compiler.

Documentation

1. EZ-USB[®] Technical Reference Manual
2. CY7C68013A/14A/15A/16A data sheet
3. GETTING STARTED WITH FXLP™ Application note
4. HIGH-SPEED USB PCB LAYOUT RECOMMENDATIONS – AN1168
5. EZ-USB FX2LP(TM)/ AT2LP(TM) RESET AND POWER CONSIDERATIONS – AN064

Go to www.cypress.com to download the latest version of the product collateral. .

Summary

This application note has introduced the reader to the differences between the FX2 and the FX2LP. As discussed in this application note, the only required change when using the FX2LP in place of the FX2 is the alteration of the crystal load capacitors and the load capacitance of the crystal. With exception of the load capacitance on the crystal, the crystal for the FX2LP contains the same requirements as that for the FX2. Additional features are also highlighted, but are defaulted to function as the FX2 would. This enables use of FX2LP in prior designs with minimal or none at all changes to the firmware. Due to the increased memory within the FX2LP, a limited number of designs may require a memory map change.

As with Cypress's FX2, the FX2LP is made available with world-class development tools and software support. Visit www.cypress.com for more device details and data sheet information.

Document History Page

Document Title: Migrating From EZ-USB FX2™ to EZ-USB FX2LP™

Document Number: 001-42079

Revision	ECN	Orig. of Change	Submission Date	Description of Change
**	1597504	SSJO/ YIS	10/11/2007	Re-catalogued application note.
*A	3149911	AASI	01/21/2011	Added content for clearer understanding on the memory map changes. Updated the application note with latest part number and collateral details.

In March of 2007, Cypress recataloged all of its Application Notes using a new documentation number and revision code. This new documentation number and revision code (001-xxxxx, beginning with rev. **), located in the footer of the document, will be used in all subsequent revisions.

EZ-USB is a registered trademark and EZ-USB FX2 and EZ-USB FX2LP are trademarks of Cypress Semiconductor Corporation. All product and company names mentioned in this document are the trademarks of their respective holders. All other trademarks or registered trademarks referenced herein are the property of their respective owners.

Cypress Semiconductor
198 Champion Court
San Jose, CA 95134-1709
Phone: 408-943-2600
Fax: 408-943-4730
<http://www.cypress.com>

© Cypress Semiconductor Corporation, 2007-2011. The information contained herein is subject to change without notice. Cypress Semiconductor Corporation assumes no responsibility for the use of any circuitry other than circuitry embodied in a Cypress product. Nor does it convey or imply any license under patent or other rights. Cypress products are not warranted nor intended to be used for medical, life support, life saving, critical control or safety applications, unless pursuant to an express written agreement with Cypress. Furthermore, Cypress does not authorize its products for use as critical components in life-support systems where a malfunction or failure may reasonably be expected to result in significant injury to the user. The inclusion of Cypress products in life-support systems application implies that the manufacturer assumes all risk of such use and in doing so indemnifies Cypress against all charges.

This Source Code (software and/or firmware) is owned by Cypress Semiconductor Corporation (Cypress) and is protected by and subject to worldwide patent protection (United States and foreign), United States copyright laws and international treaty provisions. Cypress hereby grants to licensee a personal, non-exclusive, non-transferable license to copy, use, modify, create derivative works of, and compile the Cypress Source Code and derivative works for the sole purpose of creating custom software and or firmware in support of licensee product to be used only in conjunction with a Cypress integrated circuit as specified in the applicable agreement. Any reproduction, modification, translation, compilation, or representation of this Source Code except as specified above is prohibited without the express written permission of Cypress.

Disclaimer: CYPRESS MAKES NO WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, WITH REGARD TO THIS MATERIAL, INCLUDING, BUT NOT LIMITED TO, THE IMPLIED WARRANTIES OF MERCHANTABILITY AND FITNESS FOR A PARTICULAR PURPOSE. Cypress reserves the right to make changes without further notice to the materials described herein. Cypress does not assume any liability arising out of the application or use of any product or circuit described herein. Cypress does not authorize its products for use as critical components in life-support systems where a malfunction or failure may reasonably be expected to result in significant injury to the user. The inclusion of Cypress' product in a life-support systems application implies that the manufacturer assumes all risk of such use and in doing so indemnifies Cypress against all charges.

Use may be limited by and subject to the applicable Cypress software license agreement.